Intro to Markov Chains

Linear Algebra and its Applications

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Markov Chains

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- If today is sunny, what is the probability that the day after tomorrow is sunny?
- How about the day after that?
- If the probability that the day after n nights is sunny is p, what is the probability that the day after $n+1$ nights is sunny?

Markov Chains

- A (discrete time, finite state) *Markov chain* is a stochastic process, i.e. a collection of random variables $X_1, X_2, \ldots$, each taking one of the finitely many states (values) $(s_1, s_2, \ldots s_n)$ such that the probability that $X_{t+1} = s_i$ only depends on the value of X_t.

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- For example, tomorrow's weather only depends on the weather today, (and not for example the day before)
- To know what values X takes tomorrow (X_{t+1}), we only need to know what X_t is
- Once we are in state s, it does not matter how we got there, the process forgets its history

Conditional Probability

- The quantity

$$\Pr(X_{t+1} = s_1 | X_t = s_2)$$

denotes the conditional probability that $X_{t+1} = s_1$, when we know that $X_t = s_2$.

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- For example, let $X_t = s_2$ be the event that today is sunny, and $X_{t+1} = s_1$ be the event that tomorrow is cloudy. Then, the expression $\Pr(X_{t+1} = s_1 | X_t = s_2)$ is the conditional probability that tomorrow is cloudy, given that we have observed that today is sunny.

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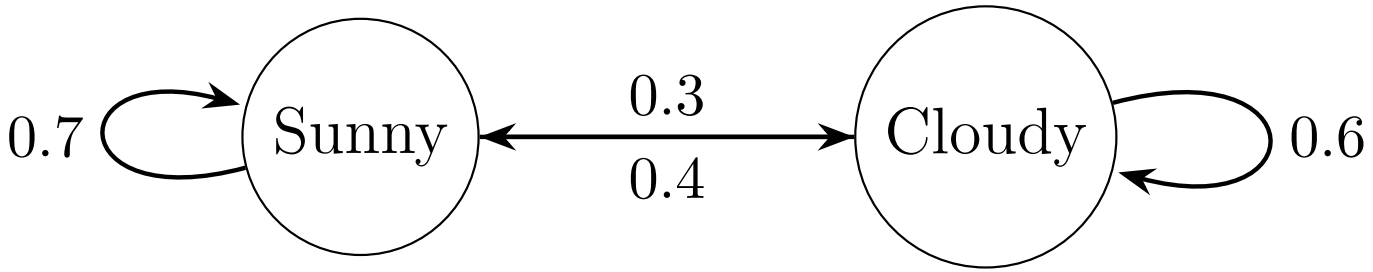
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- In a Markov chain the probabilities of the next state only depend on the current state (and not the process history), so the quantity $\Pr(X_{t+1} = s_1 | X_t = s_2)$ is independent of t.
- It is convenient to write these conditional probabilities as *transition proabilities*, $p_{ij} = \Pr(X_{t+1} = s_j | X_t = s_i)$, for $i, j = 1, \ldots, n$.

Diagram of the transition probabilities

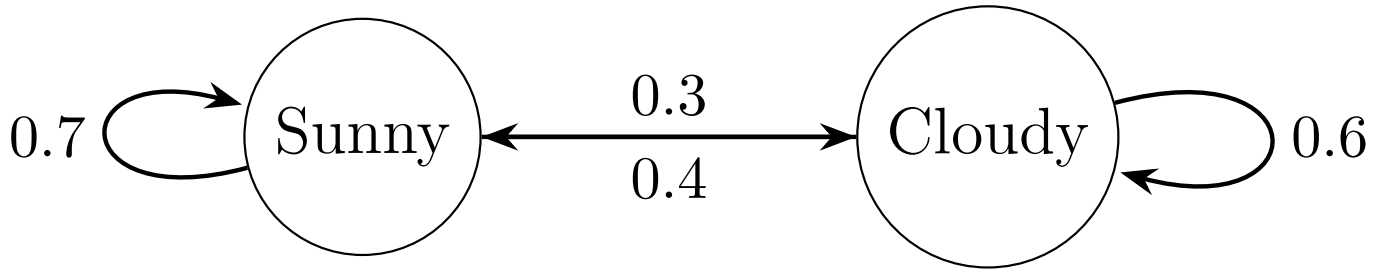
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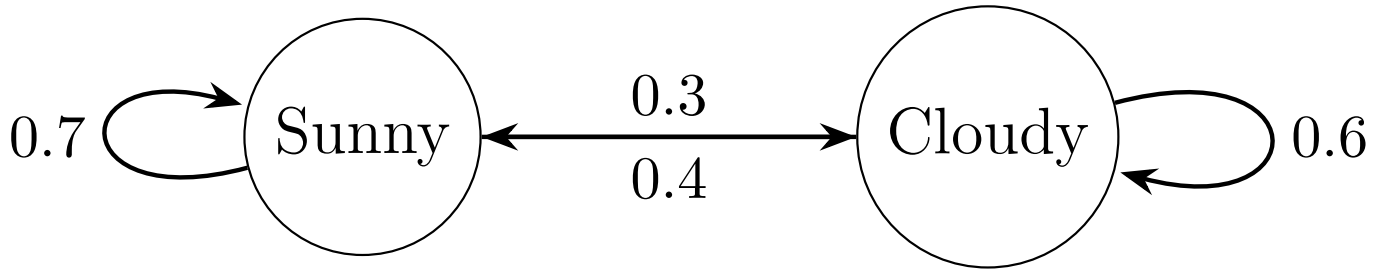
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- We can also represent the transition probabilities as the transition matrix:

$$P = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} = \begin{bmatrix} p_{1,1} & p_{2,1} \\ p_{1,2} & p_{2,2} \end{bmatrix}$$

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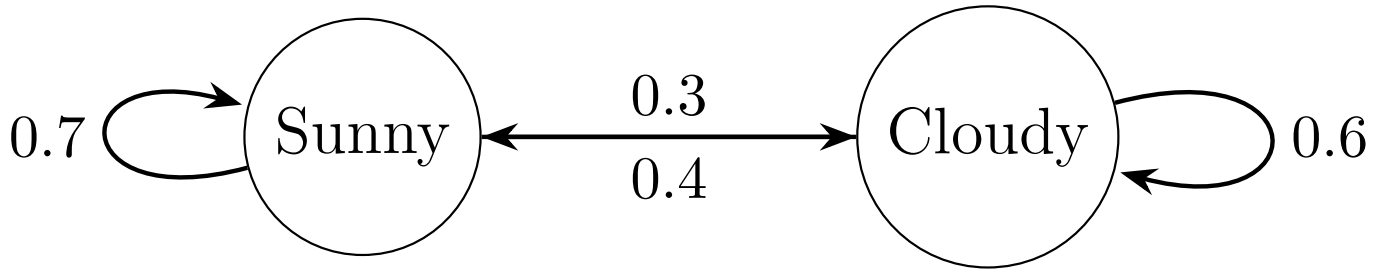
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- Example. If today is sunny, $(\Pr(X_0 = s_1) = 1)$, we have $\mu_0 = (1, 0)^T$. The next state is given as $\mu_1 = P\mu_0 = (0.7, 0.3)$.

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- The day after that is given by $P\mu_1 = PP\mu_0 = (0.61, 0.39)^T$

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$$\begin{aligned}\mu_{n+1} = P\mu_n &= (0.7p + (0.4)(1-p), 0.3p + 0.6(1-p))^T \\ &= (0.4 + 0.3p, 0.6 - 0.3p)^T.\end{aligned}$$

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- It is sunny with probability $0.4 + 0.3p$, which can be anything from 0.4 to 0.7, depending on what p is.

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- If we want to predict n steps forward, we just need to raise the matrix P to the nth power and hit it with our current state vector
- What do you think will happen to the matrix P^n, as we increase n?

```
import numpy as np
A=np.array(([0.7,0.3],[0.4,0.6]))
print(A)
[[0.7  0.4]
[0.3  0.6]]
from numpy.linalg import matrix_power
matrix_power(A,10)
array([[0.5714311,  0.5714252],
       [0.4285689,  0.4285748]])
matrix_power(A,100)
array([[0.57142857,  0.57142857],
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The matrix stabilizes, i.e., there is not much difference between P^{100} or P^{99}

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- This 0.57 is also the long term average of the sunny days
- Caveat: This interpretation is only true for *ergodic* Markov chains - more on this later

Steady-state-solution

- If the difference between P^{100} and P^{99} is small, we have that the states μ_{100} and μ_{99} are essentially the same
- A *steady-state-solution* of the chain is a state vector π that satisfies $\pi = P\pi$ - the state is steady in that it doesn't change over time
- This can be equivalently written as $(P - I)\pi = 0$, i.e., π is a right eigenvector of matrix P, with eigenvalue 1.

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- Moral: Usually we multiply with A from the left, but in the literature on Markov chains the other convention is more prevalent – you can always interpret it the other way by taking transposes

More on Eigenvectors

- The eigenvalues are found as the solution to the system

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- Hence

$$A^k x = \sum_{i=1}^{n} a_i \lambda_i^k v_i.$$

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- The matrix D is diagonal, so if $D = \text{diag}(\lambda_1, \lambda_2, \dots \lambda_n)$, then $D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots \lambda_n^k)$

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- Interpretation: If A is the transition matrix, the Perron-Frobenius eigenvector gives the steady-state solution, and this is also the unique stable state of the Markov chain that describes the long term average of the process
- This theorem also applies if $P^k = A$ for some k, i.e., if after k steps there is a positive probability of reaching any state from any other state, then the steady-state solution is unique and it is the long term average

Ergodicity

- Loosely speaking, an ergodic process is a process where "average over state space" equals "average over time"

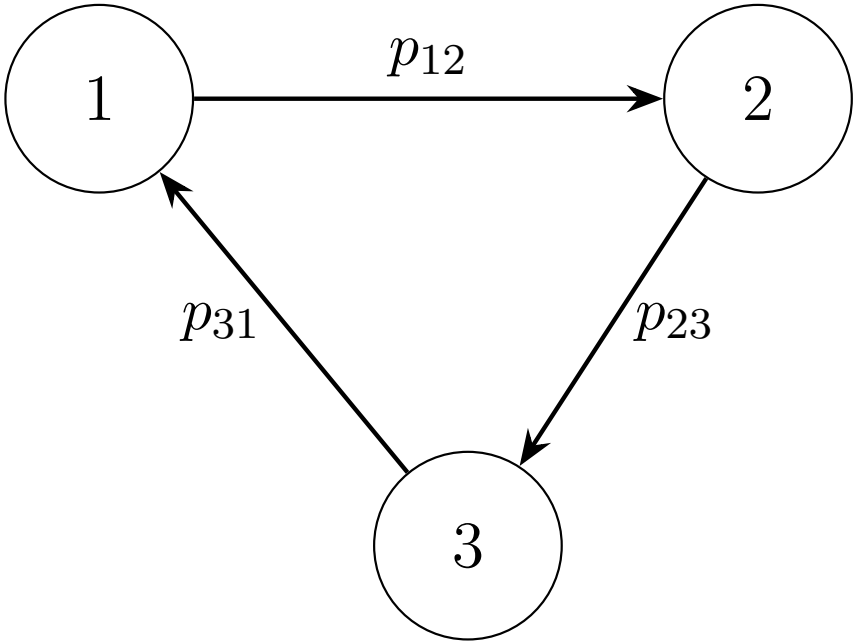
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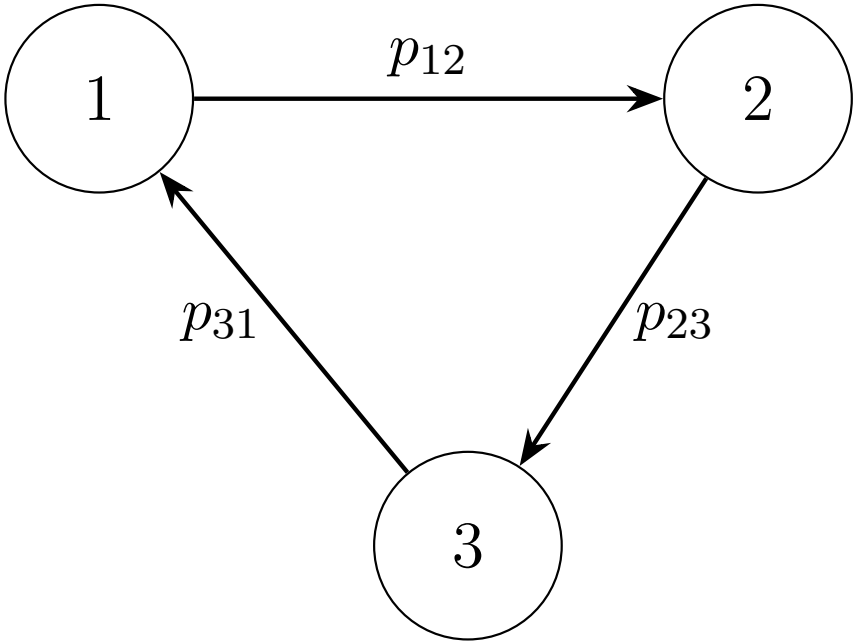
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- There are essentially two obstructions to a Markov chain being ergodic - it fails if the process is *cyclic* or *reducible*

An example of a cyclic Markov chain



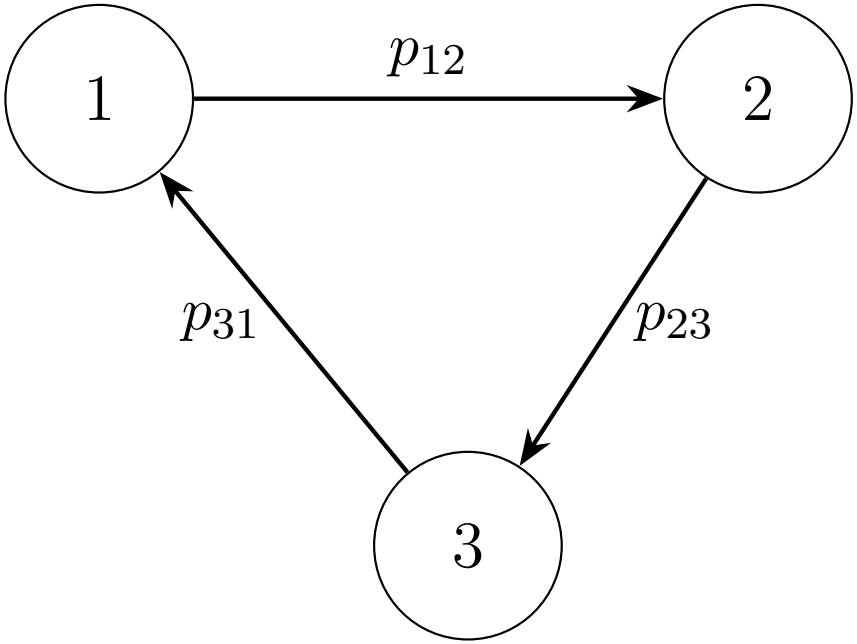
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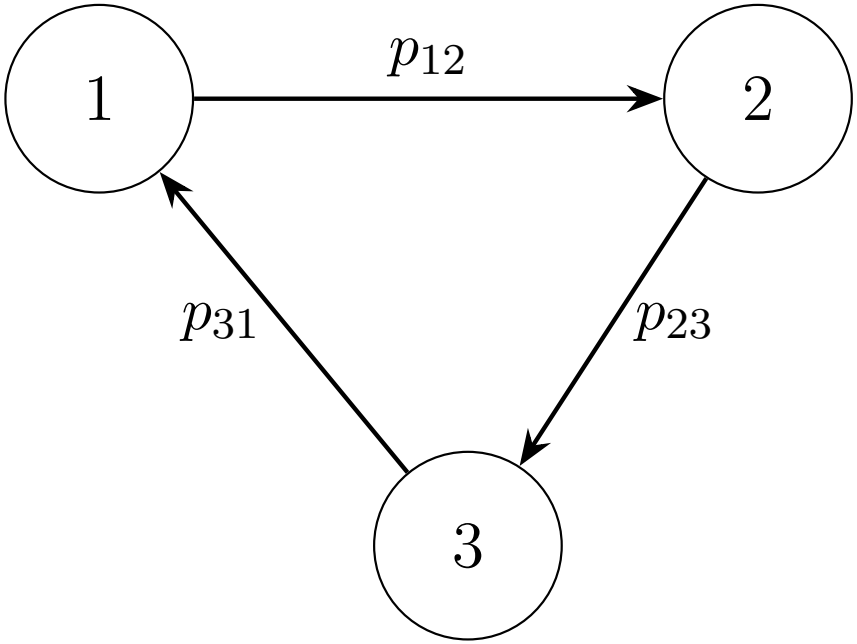
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- If we start from state 1, we end up back to state 1 every 3rd step, $P^k(1,0,0) = (1,0,0)$ for $k = 3, 6, 9, \ldots$
- Similarly $P^k(1,0,0) = (0,1,0)$ for $k = 1, 4, 7, \ldots$, and $P^k(1,0,0) = (0,0,1)$ for $k = 2, 5, 8, \ldots$

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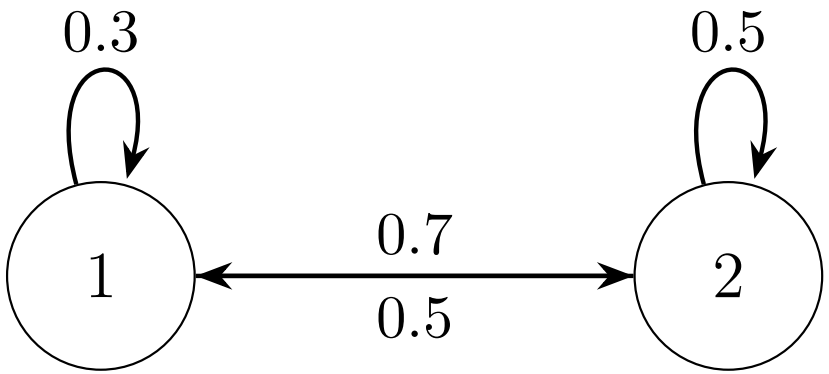
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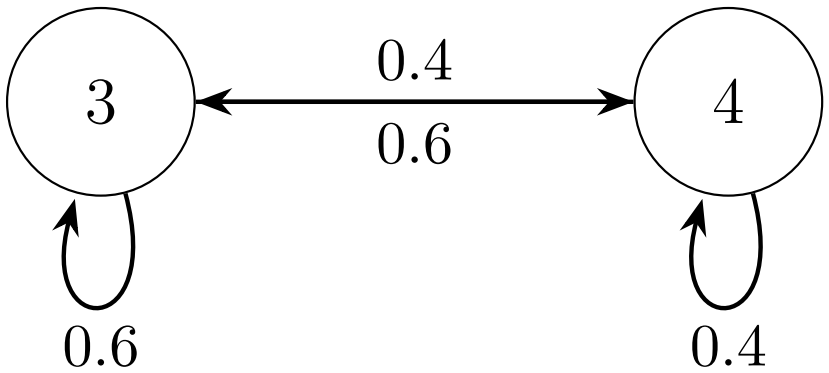
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- Similarly $P^k(1,0,0) = (0,1,0)$ for $k = 1, 4, 7, \ldots$, and $P^k(1,0,0) = (0,0,1)$ for $k = 2, 5, 8, \ldots$
- There is a steady state solution $\mu = (1/3, 1/3, 1/3)$, but this does not represent the limiting behavior
- There is no k such that P^k would have all positive entries

Reducible Markov Chain Example



Description:

- This is a **reducible** Markov chain with 4 states.
- There are no transitions between the two groups.
- Each subchain is **irreducible** on its own.
- The transition matrix is block diagonal:



$$P = \begin{bmatrix} 0.3 & 0.5 & 0 & 0 \\ 0.7 & 0.5 & 0 & 0 \\ 0 & 0 & 0.6 & 0.6 \\ 0 & 0 & 0.4 & 0.4 \end{bmatrix}$$

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- These don't describe the long term behavior, because depending on where start we get difference convergence behavior (if we start at 1, we are never going to reach 3 nor 4)

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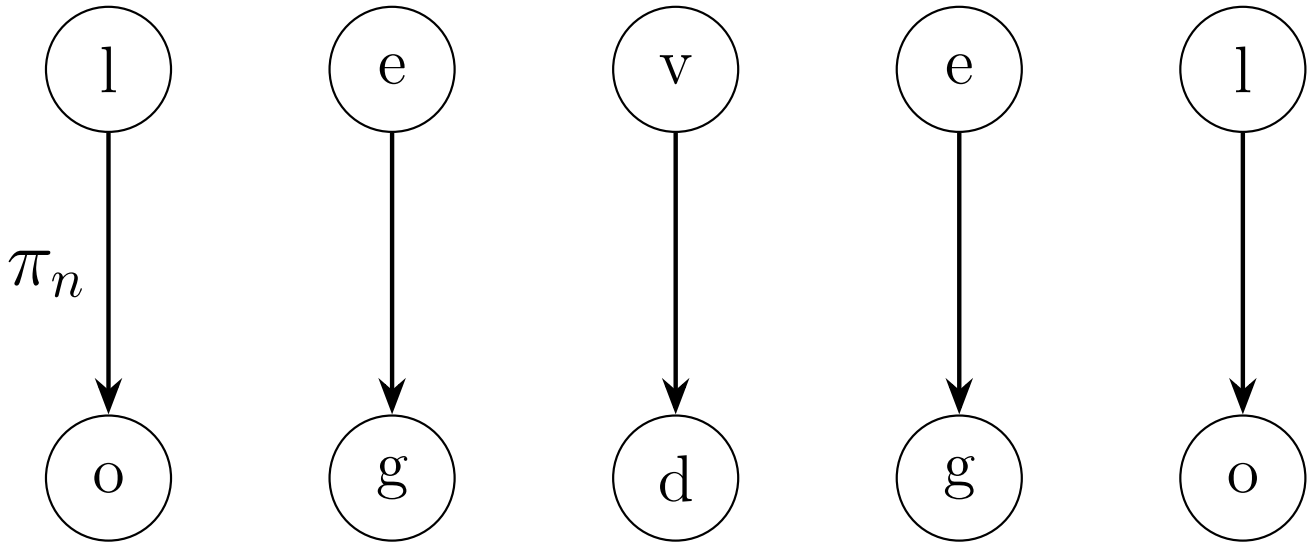
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  - By starting the chain in any state μ, and the computing $P^n\mu$ for n a high number

Application to cipher encrypting

- A substitution cipher encryption means encrypting a string message by randomly scrambling the letters in it.

Example: Original string: 'level'

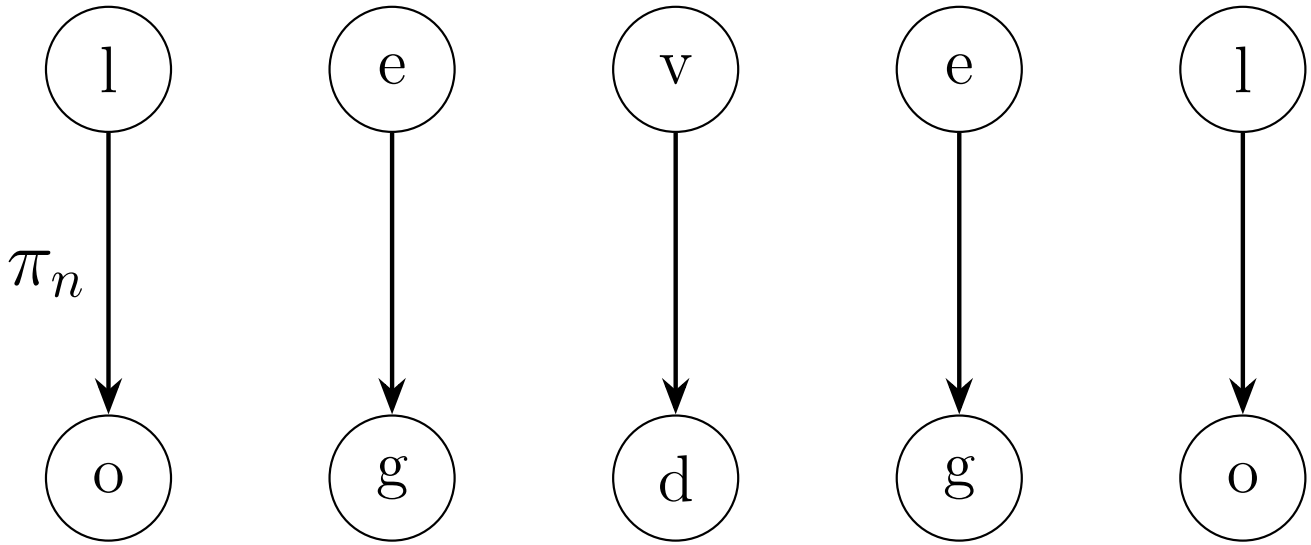


Scrambled string: 'ogdgo'

Application to cipher encrypting

- A substitution cipher encryption means encrypting a string message by randomly scrambling the letters in it.
- Given alphabet of size n and a string s, we construct an encrypted string by permuting the alphabet with map π_n:
- Goal: Crack the code (assume known alphabet, and that the message was in English)

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- Let N be the length of the message, and P_{ij} the transition probabilities estimated from a large corpus, and e the encrypted message
- Goal: Find decryption f : (a permutation of the alphabet) such that $\prod_N P_{f(i),f(j)}$ is maximized
- Problem: the space of permutations is huge! For example, if the alphabet is of size $n = 26$, the number of permutations is $26! \approx 4 \times 10^{26}$

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- Otherwise, set $f_{k+1} = f_*$ with probability $S(f_*)/S(f_k)$. Otherwise, set $f_{k+1} = f_k$
- Repeat the above many times for $k = 1, 2, \ldots M$, and keep track of the best guess

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- The algorithm described above is another Markov chain that explores the space of permutations of letters
- This is an ergodic chain, so it eventually converges to a stable distribution
- The transition probabilities are constructed in a clever way such that stable distribution is actually a very reasonable statistical estimate
- This algorithm is known as the *Metropolis-Hastings* algorithm, and it is a standard work horse in *Bayesian statistics*